

Fast, Faithful, or Diverse — Pick Two

Decoding strategy trade-offs in open-ended text generation with GPT-2

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01 HYPOTHESIS

Decoding strategy is usually treated as a minor inference setting. **Does it actually trade off faithfulness, diversity and cost in a way that matters for which strategy you should pick?**

A language model defines a distribution over the next token, but not how to turn that distribution into text. **Greedy** and **beam search** (Meister et al., 2020) pick the most probable continuation; **sampling** methods — temperature, top-p (Holtzman et al., 2020), top-k (Fan et al., 2018) — inject randomness instead. Which one is "best" depends entirely on what the task actually needs, and that's rarely made explicit.

We run one model under five decoding configurations on two tasks — summarisation (a reference exists) and open-ended continuation (it doesn't) — and measure what each strategy actually costs and buys.

H1 · Quality

Deterministic decoding should score higher on reference-based quality (ROUGE) than sampling, since matching a single reference rewards picking likely tokens, not exploring alternatives.

H2 · Diversity

Sampling should produce far more varied outputs across repeated runs of the same prompt than greedy or beam search, which are (near-) deterministic by construction.

H3 · Cost

Latency should scale with beam width — beam search should be the most expensive strategy per generation, sampling roughly as cheap as greedy.

02 METHODOLOGY

Five decoding configurations, one model

■ Greedy ■ Beam (k=4) ■ Sample T=0.7

■ Sample T=1.0, p=0.9 ■ Sample T=1.2, p=0.95

Config	Setting
■ Greedy	argmax, no sampling
■ Beam (k=4)	4 beams, early stopping
■ Sample T=0.7	temperature only
Sample T=1.0, p=0.9	temperature + nucleus
Sample T=1.2, p=0.95	higher temp + wider nucleus

Model: GPT-2 (124M, frozen, off the shelf) — open weights, no API key, no fine-tuning. Every config uses the exact same model and prompts; only the decoding call changes, so any difference in outcome is attributable to decoding, not the model.

One dataset, two roles. CNN/DailyMail (Hermann et al., 2015; See et al., 2017) supplies both tasks: the article's lead sentence seeds an open-ended *continuation* prompt (no reference, diversity is the point), and the full article paired with its human-written highlight is a *summarisation* pair (has a reference, faithfulness is the point).

Data & protocol

Experiment	Prompts	Samples/prompt
H1 · Quality (summarisation)	60	1
H2 · Diversity (continuation)	60	5
H3 · Cost (continuation, timed)	10	3 repeats

60 prompts × 5 configs = **300 generations for H1**; 60 × 5 × 5 = **1,500 for H2**. Metrics: ROUGE-1/2/L against the real reference (Lin, 2004), distinct-2 across the 5 resamples per prompt (fraction of unique bigrams — higher means more varied outputs), and n-gram repetition rate within a single output.

03 RESULTS

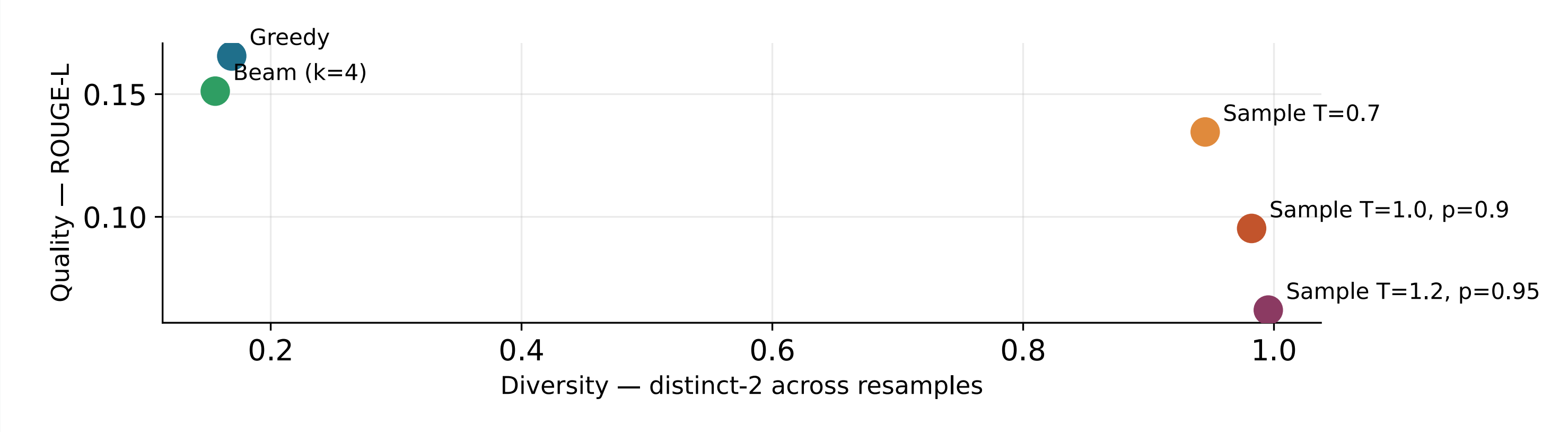


Fig 1 · H1 & H2. Mean ROUGE-L against a real reference (y) vs. mean distinct-2 across 5 resamples (x), one point per decoding config, n=60 prompts.

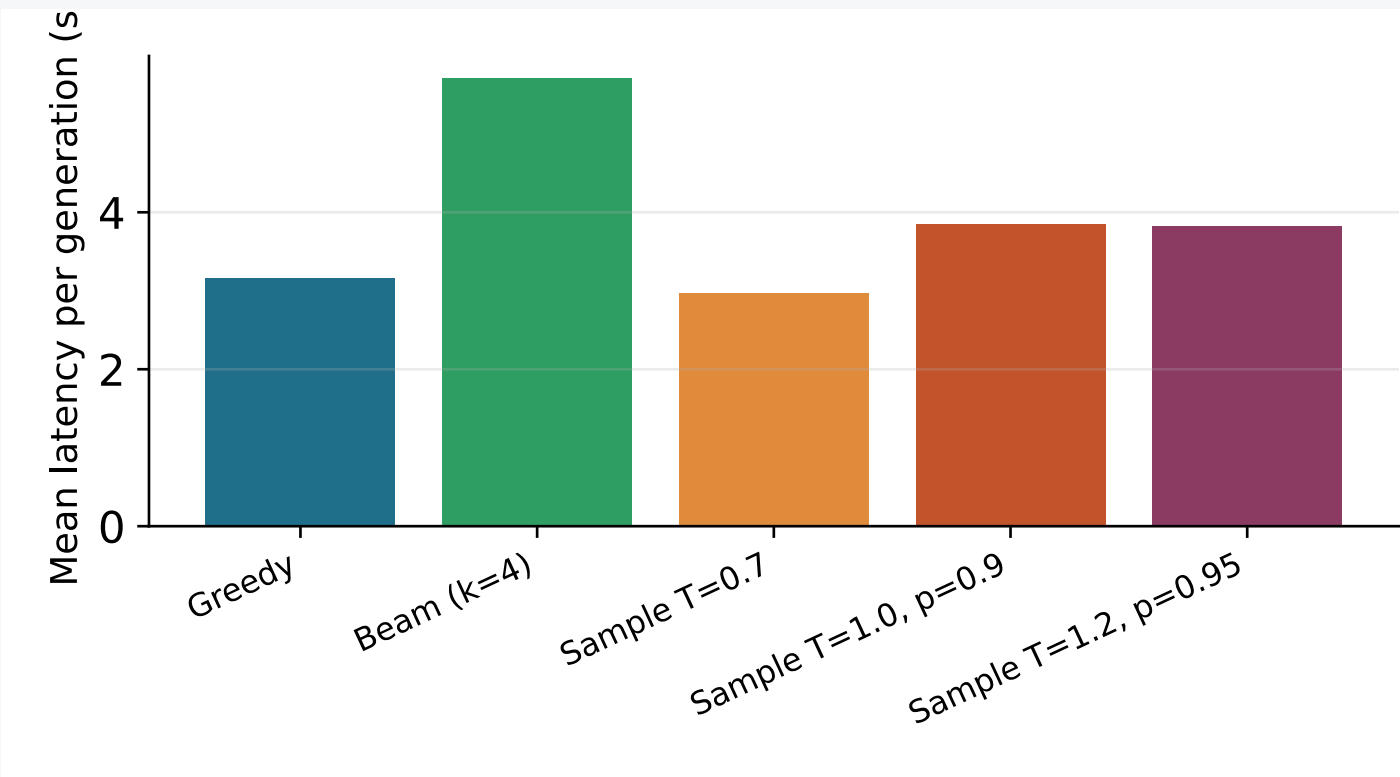


Fig 2 · H3. Mean wall-clock latency per generation (n=10 prompts × 3 repeats), same prompt and output length across configs.

Fig 3 · one prompt, five strategies

Greedy	same output on all 5 runs
Beam(4)	same output on all 5 runs
Sampling	5 runs, 5 different stories

Greedy/beam are deterministic — all 5 resamples word-for-word identical; sampling diverges every time. Full text: results/exp2_examples.json.

Findings

H1 confirmed, with a twist: deterministic decoding beats sampling on ROUGE-L (greedy 0.166, beam 0.151, vs. 0.135 → 0.062 as sampling loosens) — but beam only wins that search over greedy on ROUGE-2 (0.058 vs. 0.051); on ROUGE-1/L greedy edges it out, echoing Meister et al.'s point that more search doesn't guarantee a better match to one reference. **H2 strongly confirmed:** distinct-2 is 0.16–0.17 for greedy/beam vs. 0.95–0.995 for sampling, scaling cleanly with temperature. **H3 confirmed:** beam is the slowest by far (5.71s vs. 2.96–3.84s), lowest throughput (7.2 vs. 10–13 tok/s).

Limitations

GPT-2-124M is small and not fine-tuned, so absolute ROUGE is low throughout — the comparison is relative between strategies, not against the state of the art. ROUGE itself is a surface-overlap metric the lecture (Notes 09) flags as unreliable for open-ended text. Single model, one dataset.